

Screenshots of Serial Monitor:

Here are both of the gestures being successfully found by the model.

```
hi: 0.421404  
sup: 0.578596  
  
hi: 0.280458  
sup: 0.719542  
  
hi: 0.672662  
sup: 0.327338  
  
hi: 0.723832  
sup: 0.276168  
  
hi: 0.570330  
sup: 0.429670  
  
hi: 0.779183  
sup: 0.220817
```

Python Editing:

In my python notebook I changed the compiler line from what is commented out to the new compiler line.

```
#model.compile(optimizer='rmsprop', loss='mse', metrics=['mae'])  
model.compile(optimizer='rmsprop', loss='categorical_crossentropy', metrics=['accuracy'])
```

The two changes made were changing the loss to categorical crossentropy loss and the evaluation metric to accuracy. Beforehand, the loss was MSE (mean squared error) and the metric was MAE (mean absolute error). These work really well for regression tasks, but not as well for classification, which is what this task was. The model's inputs had been one hot encoded and the task of the model was binary classification. This means that it was choosing between two outputs. MAE was designed for regression tasks, meaning the model is finding a value to output. Therefore we changed the metric to accuracy since that is the traditional and best choice for classification. We also changed the loss to categorical cross entropy since it is optimized for OHE with classification. Although MSE would work, it is better optimized for regression and would give us worse results.

To fix everything after that we changed the graph's label from MAE to accuracy.

```
] : METRIC_TO_PLOT = "accuracy" # Update this if you changed the metric used during training.
```